

Jason Duran Dutton

Independent Researcher | Physics, Photonics, and Systems Engineering

AMO / Optical Properties of Materials

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Research manuscripts | Photonic QEC engine

Profile

Independent theoretical physics researcher with more than 15 years of self-directed study, manuscript development, and computational model building. My work connects geometric descriptions of electron and photon behavior with optical confinement, polarization, stability, and photonic error correction. I am particularly effective at turning difficult physics into precise, intuitive verbal explanations and at evaluating whether an explanation is physically coherent.

AMO and Optical-Physics Focus

- **Atomic structure and transitions:** qualitative reasoning about hyperfine structure, polarization, selection-rule intuition, and the distinction between σ and π transitions.
- **Optical properties of materials:** refractive-index contrast, waveguide geometry, mode confinement, photonic stability, and the relationship between material response and field propagation.
- **Polarizability and external fields:** geometric interpretation of induced displacement, field response, and stability under perturbation.
- **Photonic systems:** SPDC, photon identity, CHSH/Bell behavior, displacement geometry, stabilizer logic, and geometric quantum error correction.
- **Model critique and explanation:** translating between physical intuition, mathematical structure, assumptions, and experimentally meaningful predictions.

Foundational and Theoretical Physics Focus

- **Geometric field modeling:** worldline geometry, curvature, torsion, invariants, field stability, and relationships between local structure and global behavior.
- **Quantum foundations:** electron/positron identity, interference, measurement, photon identity, SPDC, Bell/CHSH correlations, and interpretation of quantum state behavior.
- **Unified theoretical frameworks:** development of conceptual operators and layered models connecting spacetime geometry, identity-space structure, dynamical invariants, and physical interpretation.
- **Scientific reasoning:** separating assumptions, mathematical structure, physical intuition, predictions, and possible tests when analyzing a theory or technical explanation.

Computational, Systems, and Applied Research Focus

- **Scientific software and modeling:** Python-based research tools, modular simulation workflows, geometric state representation, error propagation, and reproducible technical documentation.
- **Quantum information systems:** geometric stabilizers, syndrome structure, displacement fields, coherence preservation, photonic encoding, and quantum error-correction concepts.
- **Industrial research translation:** converting research objectives into prints, controls, ladder logic, instrumentation, commissioning procedures, field tests, and operational decisions.
- **Reliability and safety engineering:** fault isolation, system integration, verification, regulatory awareness, high-consequence troubleshooting, and communication across technical disciplines.

Selected Research

Independent Theoretical Researcher

2010–Present | Castlewood, VA

- Developed a unified geometric research program spanning worldline models, identity-space invariants, photonic stability, and quantum error correction.
- Built the **SEFI_QEC_Stack**, a photonic quantum error-correction engine for geometric stabilizers, syndrome structure, displacement fields, and correction mapping.
- Developed and documented interpretations of electron/positron identity, interference, measurement, SPDC, CHSH correlations, polarization evolution, and field stability.
- Prepared four research manuscripts for external review and made the supporting research portfolio publicly available for inspection.
- Refined physics explanations through iterative dialogue with AI systems, with emphasis on conceptual consistency, clear assumptions, and generalization across problems.

Research Portfolio

- **Geometric Worldline Foundations of the Electron Field** (JMP26-AR-02470): electron/positron identity, interference, and measurement.
- **SEFI: The Single Entity Field Interpretation** (JMP26-AR-02501): layered identity-space geometry and invariants.
- **Unified Geometric Field Theory from Worldline and Identity-Space Invariants** (JMP26-AR-02666): spacetime geometry and dynamical invariants.
- **Photonic Quantum Error Correction from Geometric Stability Principles** (ADV26-AR-04574): stability surfaces, displacement geometry, and correction mapping.

Technical and Communication Strengths

Physics: AMO concepts | optical response | photonics | quantum foundations | geometric modeling

Computing: Python-based simulation workflows | modular research software | GitHub | LaTeX

Communication: spoken technical explanation | analogy with explicit limits | assumption checking | critique of model outputs

Professional Engineering Experience

Lead Electrician – Pilot Process Research Installation Mineral Refining Company / Lone Mountain Refining | 2015–2016

St. Charles, Virginia | Virginia Tech-directed pilot deployment of patented Hydrophobic–Hydrophilic Separation (HHS) technology | mineralsrefining.com

- Served as the first electrician on site and led full electrical integration for a pilot process designed to recover ultra-fine coal and metallic-ore particles from preparation-plant waste streams.
- Owned field coordination of control prints, ladder logic, sensor interfaces, system-level electrical decisions, installation verification, commissioning, and safety compliance.
- Integrated and tested phase-separation vessels, vibrating mixers, candle filters, pentane-based hydrophobic-liquid recovery loops, and advanced dewatering modules.
- Supported more than 1,200 hours of field testing under MSHA and DMME oversight, including validation of ultra-fine coal recovery below 45 micrometers and single-digit moisture control.
- Contributed to pilot results reporting feed-ash reduction from 40–67% to 3–5%, combustible recovery up to 97%, and HHS applicability to metallic-ore particles below 10 micrometers and 1 micrometer.
- Supported environmental and economic outcomes including extended slurry-pond life, reduced waste deposition, improved plant economics, and recovery of previously impounded fines.

Underground Electrical Repairman Paramount Contura / Alpha Metallurgical Resources | 2022–Present

Safety-critical diagnostics and maintenance requiring disciplined troubleshooting, reliability-first decisions, and clear communication under operational constraints.

Underground Electrician / Maintenance Foreman

Alpha Natural Resources | 2010–2016

Electrical systems maintenance, fault isolation, and safety compliance in a highly regulated industrial environment.

Fabricator, Welder, and Foreman

Dominion Steel | 2016–2018

Led shop and field work while coordinating system interpretation, quality execution, and production deadlines.

Education and Qualification Context

Castlewood High School – High School Diploma, 2002

Mountain Empire Community College – Underground and surface mining electrical study

Brevard Community College / UCF – CIS and business-management coursework

My physics preparation is an independent, sustained research program rather than an awarded PhD; the linked manuscripts and software provide the most direct evidence of my current research level and technical scope.